

Biofouling in industrial equipment: a look at quorum sensing and quorum quenching as anti-fouling strategies in membrane bioreactors

Anusha Yethadka Ganaraja & Lavanya Mulky

To cite this article: Anusha Yethadka Ganaraja & Lavanya Mulky (2024) Biofouling in industrial equipment: a look at quorum sensing and quorum quenching as anti-fouling strategies in membrane bioreactors, Environmental Technology Reviews, 13:1, 235-250, DOI: [10.1080/21622515.2024.2317819](https://doi.org/10.1080/21622515.2024.2317819)

To link to this article: <https://doi.org/10.1080/21622515.2024.2317819>



© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group



Published online: 21 Feb 2024.



[Submit your article to this journal](#)



Article views: 1557



[View related articles](#)



[View Crossmark data](#)



Citing articles: 3 [View citing articles](#)

REVIEW ARTICLE



Biofouling in industrial equipment: a look at quorum sensing and quorum quenching as anti-fouling strategies in membrane bioreactors

Anusha Yethadka Ganaraja and Lavanya Mulky

Department of Chemical Engineering, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, India

ABSTRACT

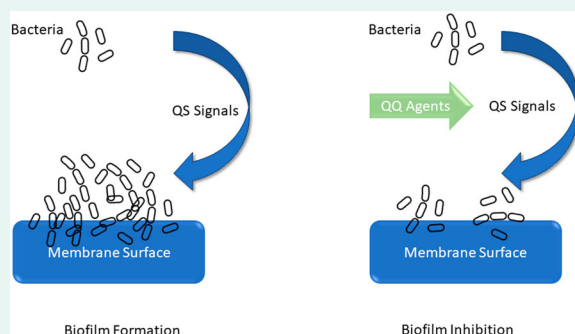
One of the major techniques used in various industrial processes and separation techniques is membrane technology, among which membrane bioreactors (MBRs) are a key component. While the demand for membrane technology is rising, membrane fouling remains a major issue. Fouling has a lot of economic consequences, and overtime proves very harmful to the environment and industry alike. Particularly in the marine and membrane bioreactor industry, biofouling is very predominant. Hence, an understanding of fouling and its mitigation is crucial to enable effective management and control of fouling in industries, minimizing damage as much as possible. With biofouling, quorum sensing and quorum quenching have come up as a novel method to monitor and mitigate biofouling. The bacterial communication through signalling molecules known as quorum sensing has gained importance because of its applicability in various fields of study. Quorum sensing regulates numerous bacterial activities such as antibiotic resistance biofilm formation, etc. Microbial communications are responsible for this biofouling behaviour. Novel techniques for stopping this communication, known as quorum quenching (QQ), appear to be effective for biofouling mitigation. This review attempts to look at quorum sensing and quorum quenching as techniques to mitigate fouling.

ARTICLE HISTORY

Received 18 January 2023
Accepted 28 January 2024

KEYWORDS

Quorum sensing; quorum quenching; biofouling; membrane bioreactor



Introduction

Over the past two decades, a major technique used in various industrial processes and separation techniques is membrane technology. It has been used as pressure-driven membranes for ultrafiltration [1], microfiltration [2] nanofiltration [3] reverse osmosis [4] and forward osmosis [5] along with other major techniques using membrane bioreactors and membrane distillation. However, while the demand for membrane technology is rising, membrane fouling

remains a major issue. Membrane fouling causes a lot of permeability issues with respect to the membrane, thereby reducing the membrane's efficiency by a lot.

Membrane bioreactor (MBR), a combination of membrane unit and the biological treatment, profoundly enhancing the water quality. For the last two decades, the Membrane bioreactor has gained significant attention for wastewater treatment and energy recovery from biomass because of its high removal rate of biochemical oxygen demand (BOD)

CONTACT Lavanya Mulky ✉ lavanya.m@manipal.edu 📍 Department of Chemical Engineering, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, 576104 Karnataka, India

© 2024 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (<http://creativecommons.org/licenses/by-nc-nd/4.0/>), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

and chemical oxygen demand (COD) [6]. Hence, it is extensively being applied to domestic and industrial wastewater treatment. Membrane fouling instigated either by cake layer formation or by pore blockage is a critical issue in membrane treatment processes [7]. Fouling continues to be the basic obstacle hindering the commercialization of membrane-based systems, as it leads to a decline in water flow rates [8]. Despite notable advancements in the widespread adoption of MBR technology in recent years, membrane fouling continues to be a major challenge due to its adverse impact on permeate flux and on process efficiency [9]

Although MBR technology offers numerous advantages, including high-quality treated water, the ability to handle a large biomass concentration, and a reduced system size, it is still constrained by the persistent problem of membrane fouling, which is considered the most significant drawback in terms of process efficiency. Fouling results in a decrease in permeate flow, leading to shorter intervals between membrane cleaning and replacement, ultimately resulting in higher operational costs [10]

Exploring novel biological strategies to mitigate membrane biofouling is of significant value in order to allow sustainable performance of membrane systems for water and wastewater treatment and other applications. This review describes recent information regarding the signal molecules and QQ strategies for control of biofouling. Also, the application of various techniques developed for fouling mitigation were assessed. This paper also comprehensively discusses the significance, impact, challenges of the membrane fouling. Membrane fouling, a major restriction in the practical application, its mechanism and antifouling strategies are briefly discussed.

Impact on the industries

Biofouling is recognized as a leading factor responsible for over 45% of total membrane fouling. It has been identified as a significant challenge, particularly in the context of nanofiltration and reverse osmosis membrane filtration processes [11]. Among the four distinct categories of fouling, biofouling is widely acknowledged as the most troublesome and problematic [12]. Few other industries affected by biofouling are depicted in the Figure 1.

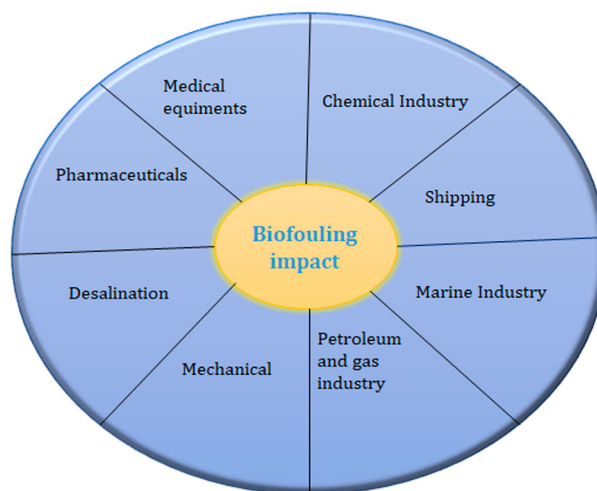


Figure 1. Schematic of various industries affected by biofouling.

Biofouling impact on membrane system performance can be outlined as follows:

- (1) Reduced membrane permeability caused by bacterial attachment and subsequent growth on the membrane surface.
- (2) Formation of calcium carbonate deposits at higher pH levels.
- (3) Build up of non-living particles due to microorganisms adhering to the membrane or enzymatic actions affecting the membrane or bonding lines.
- (4) Intensified concentration polarization due to the presence of biofilms on the membrane surface.

[Biofouling and control approaches in membrane bioreactors]

Fouling detrimentally impacts plant efficiency, thus affecting its economic equilibrium [13].

The potential of marine renewable energy devices to facilitate the establishment and spread of non-native species due to biofouling has been recognized as a significant environmental concern by scholars and policymakers. Reducing or preventing the accumulation of biofouling is a pivotal managerial approach to mitigate adverse technical and environmental effects on marine energy devices. Biofouling can lead to technical consequences for the operational efficiency of wave and tidal devices. This includes the potential to enlarge the effective size of components, like turbine blades, thereby causing heightened drag and inertia loads. Moreover, it can elevate surface roughness and irregularities,

Table 1. Consequence of various fouling types in different industries.

Fouling type	Industry/process	Consequence	Reference
Biofouling	Shellfish aquaculture industry	Increased labour costs and reduced value of product.	Biofouling and the shellfish industry
Biofouling	Shellfish aquaculture	Physical damage, mechanical interference, biological competition and environmental modification, while infrastructure is also impacted	The impact and control of biofouling in marine aquaculture: a review
Biofouling	Membrane distillation	Reduces the vapour pressure driving force by hindering mass and heat transport, consequently enhancing temperature and salt concentration at the membrane surface.	Membrane distillation biofouling: impact of feedwater temperature on biofilm characteristics and membrane performance
Biofouling, colloidal, inorganic, organic	Spiral wound membrane systems	Performance of the modules is affected	Review on strategies for biofouling mitigation in spiral wound membrane systems
Biofouling	Feed spacer	Performance decline in membrane-based water filtration system	Predicting the impact of feed spacer modification on biofouling by hydraulic characterization and biofouling studies in membrane fouling simulators
Biofouling	Membrane bioreactor	Severe flux decline, high energy consumption, and frequent membrane cleaning or replacement	Membrane biofouling in pilot-scale membrane bioreactors (MBRs) treating municipal wastewater: impact of biofilm formation
Biofouling	Reverse osmosis membranes	Formation of an additional 'barrier' formed by extracellular polymeric substances (EPS) reduces salt mass transport through the fouled membrane.	Impact of microfiltration treatment of secondary wastewater effluent on biofouling of reverse osmosis membranes
Biofouling	Reverse osmosis membranes	Decreased salt rejection and permeate flux	Role of extracellular polymeric substances (EPS) in biofouling of reverse osmosis membranes
Biofouling	Reverse osmosis membranes	Sequential decline of Reverse osmosis membrane performance indicators (feed channel pressure drop, permeability and salt rejection)	Impact of membrane biofouling in the sequential development of performance indicators: feed channel pressure drop, permeability, and salt rejection
Biofouling	Polyamide membranes	Degrade amide bonds	Yamano, N., Nakayama, A., Kawasaki, N., Yamamoto, N., Aiba, S., 2008. Mechanism and characterization of polyamide 4 degradation by pseudomonas sp. J. Pol. Environ. 16(2), 141–146. http://dx.doi.org/10.1007/s10924-008-0090-y .
Biofouling	Aquafarms	Reduce the efficiency of oxygen transfer in aquafarms	Mann, K.V. 2011. Theoretical perspectives in medical education: past experience and future possibilities. Med Educ., 45(1):60-80

leading to modifications in the flow dynamics around components and resulting in changes to lift and drag coefficients. These augmented drag and inertia effects may result in decreased energy extraction efficiency for certain Marine Renewable Energy designs [14]. Concerning marine coatings, microbial proteins from marine organisms adhering to hull surfaces create a range of issues, including resistance and loss. In the fields of biomedicine and food separation, membrane fouling often results in challenging problems, including flux reduction and rejection rate issues [15].

Saudi Arabia plays a significant role in global desalinated water production, contributing approximately one-third of the world's capacity. Current desalination technologies in the Kingdom encompass the multi-stage flash method and the reverse osmosis process. The Reverse Osmosis (RO) process is preferred due to its simplicity, cost-effectiveness, and ease of maintenance. However, recent critical challenges associated with RO membrane processes include fouling, biofouling, and biocorrosion [16]

Marine biofouling prevention is a worldwide industry of five billion US dollars a year, and growing [17]. Table 1 depicts the consequence of various fouling types in different industries.

When the membrane surface is covered (internally or externally) due to accumulation of substances, it is termed as membrane fouling and when this is caused due to the attachment of microorganisms on surfaces or other bacterial activity, it is termed as biofouling [18]. This bacterial adhesion on surfaces results from complex chemical communications between bacteria in the environment and the chemical, biological and physical characteristics of the surface and the microbial cells. These organisms may also further lead to corrosion of equipment, particularly those of underwater structures. Biofouling is an inevitable phenomenon that has led to a lot of economic loss, and while various studies have been conducted to mitigate it [19], they have several limitations. A novel method proposed in the recent decade has been the study and inhibition of

quorum sensing (QS), to treat biofouling with the least economic and environmental harm possible.

Significance of membrane technologies in various fields

Membrane technologies can be applied to extract nutrients and facilitate the crystallization of struvite from the anaerobic digestion supernatant. This supernatant is particularly suitable for nutrient recovery in wastewater treatment plants. The use of membrane systems presents an intriguing avenue for efficiently recuperating nutrients from the supernatant, thereby supporting the widespread adoption of these technologies [20]. Over time, significant quantities of oily wastewater resulting from petrochemical extraction and transportation have been released into the environment, leading to severe damage in local areas. Canada faces substantial oil pollution risks, given that approximately 180 vessels transporting over 80 million tonnes of oil navigate its extensive 243,000 km coastline. The efficacy of membrane technology in treating oily wastewater across diverse industries makes it a promising option for the on-site treatment of wastewater generated during response operations [21]. Hence, due to its track record, membrane technology stands as a strong contender for addressing the treatment needs of the wastewater generated in response operations [22].

The consumption of pathogen-contaminated water has tragically led to loss of life. This situation underscores the immediate necessity for research approaches that can prevent, treat, and eliminate harmful pathogens present in wastewater. Consequently, ensuring effective water treatment has become a matter of paramount importance. Membrane technology offers a means to achieve purer, cleaner, and pathogen-free water through the separation of water using a permeable membrane [23]. Membrane technology finds applications in the large-scale industrial production of microalgal biomass [24]. Significant gaseous pollutants like CO₂, SO₂, and NO_x are emitted during fossil fuel combustion, energy-intensive material production, and waste processing. Industry and research communities are collaboratively working to enhance their capture and reduce emissions. The ultimate objective is to adhere to stricter emission limits, a task that's becoming more challenging with conventional techniques. Notably, membrane separation technology

holds substantial promise in environmental protection applications. It can effectively complement other separation methods and generally consumes significantly less energy [25]. Membrane technologies have been extensively employed for various gas/liquid or gas/gas separations across a wide array of industrial contexts. These include municipal and industrial wastewater treatment, producing N₂ from air, recovering H₂ from ammonia production setups, and capturing vapours from processed gas streams, among others. Particularly in recent times, focused research has broadened the scope of gas separations using membranes [26].

Economic significance and environmental impact

A study conducted in the Netherlands to assess the cost of fouling in full-scale reverse osmosis and nanofiltration installations revealed that fouling costs accounted for approximately 11% and 24% of operating expenses in nanofiltration and reverse osmosis plants, respectively [27]. The impact of biofouling on aeration devices escalates production costs due to heightened electricity consumption, increased maintenance expenses, and labour costs. In instances of heavy fouling, paddle wheel power consumption can surge by up to 50%. Some farms have reported an average 20% increase in power consumption and 50% higher maintenance and repair costs due to fouling [28]. Low plant availability and increased chemical costs due to high chemical cleaning frequency and the use of biocides [29].

Biofouling: an overview

Biofouling is currently one of the major problems faced by several industries [30] such as that the paper and pulp industry, oil and natural gas industry, power generation industry, food processing industry, wastewater treatment industry, etc. Owing to the growing awareness on the topic, interest and studies conducted on it has only risen in the past decade, as shown in Figure 2.

As mentioned above, biofouling (also called biological fouling) is the accumulation and growth of microbes or other microbial species onto the membrane surface, causing membrane fouling [31]. This microbial layer is termed as biofilm, and is on the membrane surface as a complex layer of polymeric matrix [32]. Biofouling causes reduced efficiency in

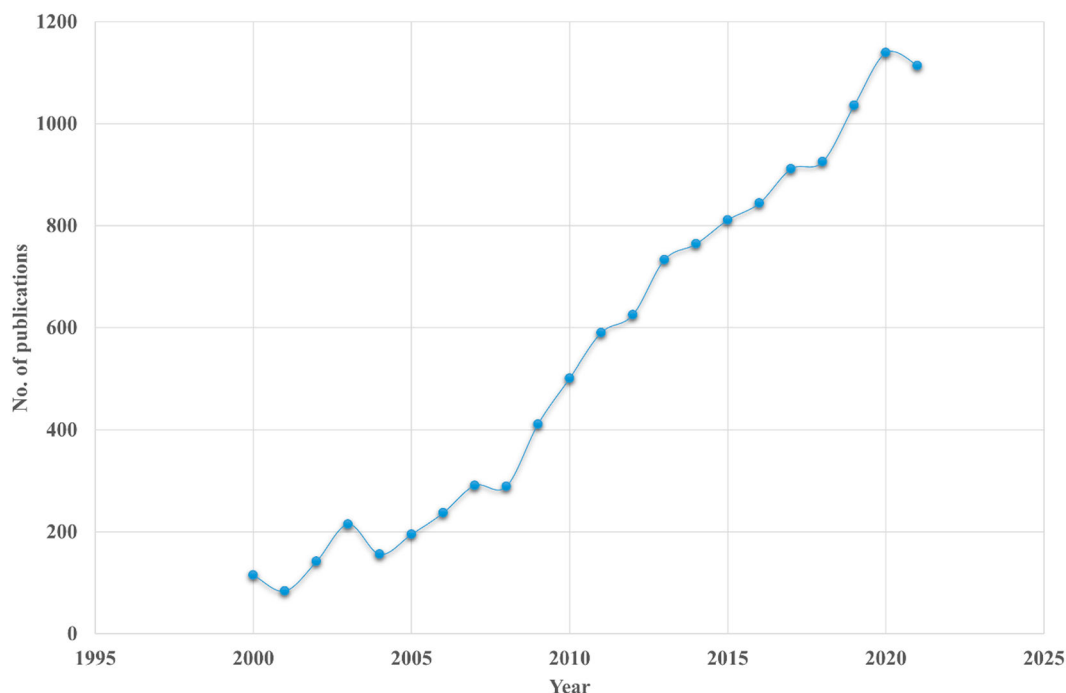


Figure 2. Number of publications on biofouling versus the year of publication in the past decade. Data taken from SCOPUS with keyword 'BIOFOULING', accessed on 15 February 2022

membrane processes by reducing the flux and separation decay. This also causes pore wetting and reduction in the membrane lifetime [33].

In biofouling, the first stage begins with the adhering of biofoulants, such as algae or bacteria, onto the surface using weak van der Waals forces. In case of irreversible attachment, microorganisms use cell adhesion structures to anchor themselves permanently. For example, Linden Harris et al. [34]

conducted a study where it was noted that development of an organic layer on the membrane surface constitutes biofouling growth, with micro fouling due to colonies of unicellular algae, and macrofouling due to multicellular algae. The arrival of more cells marks the beginning of stage 2, which is facilitated by the first colonist by reserving more diverse adhesion sites (hosting centres) and holding the biofilm together by building the matrix. Stage 3,

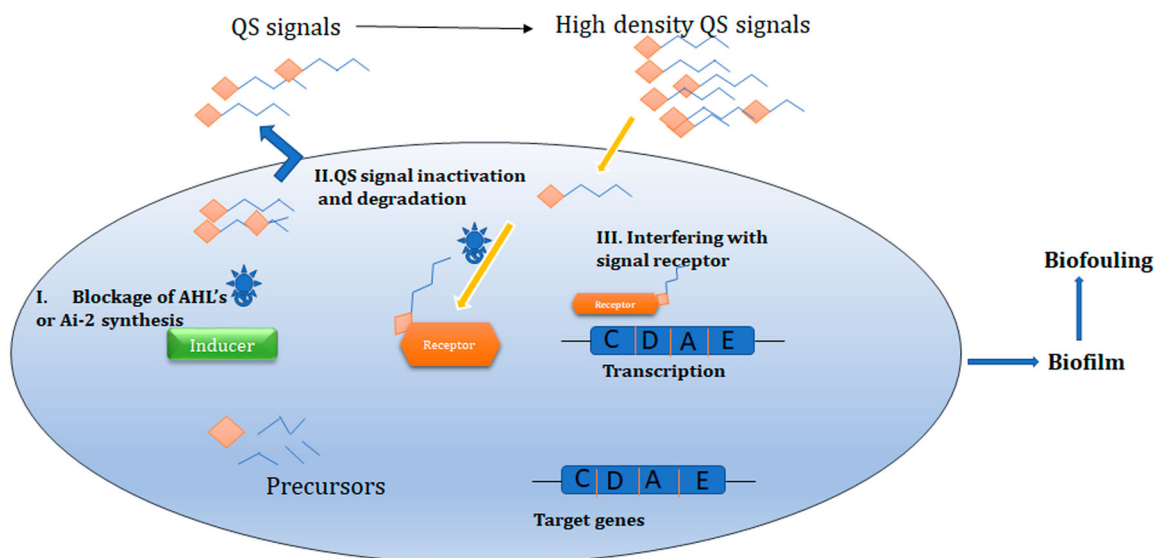


Figure 3. Illustration of probable QQ strategies for control of biofouling.

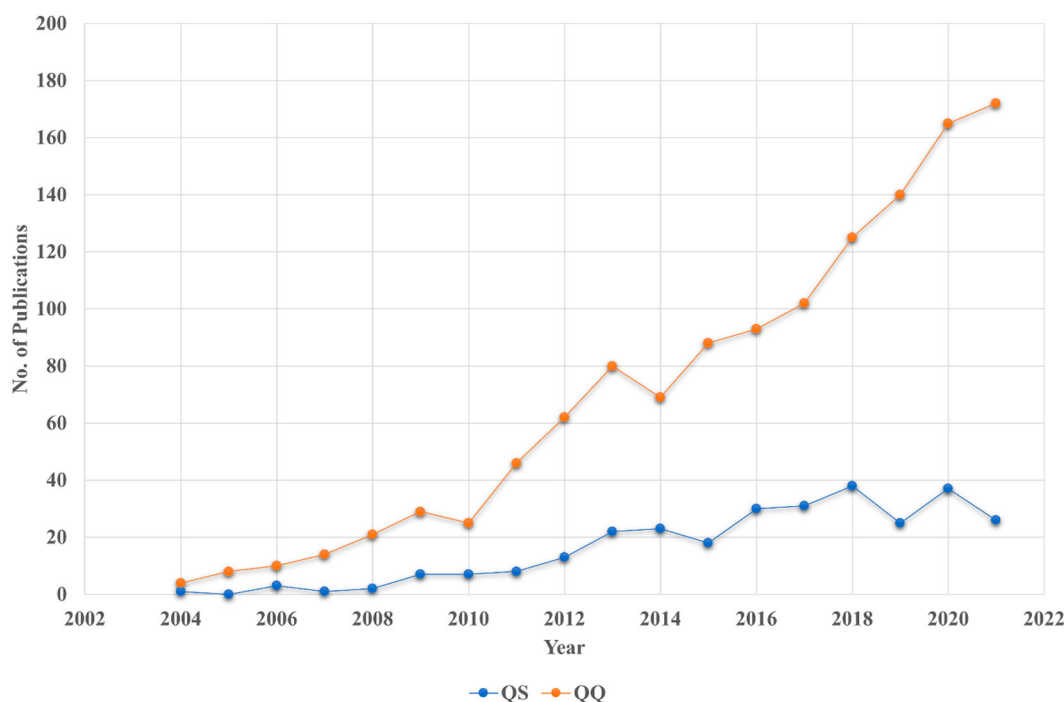


Figure 4. Number of publications on quorum sensing (in blue) and quorum quenching (in orange) versus the year of publication in the past decade. Data taken from SCOPUS with keywords 'QUORUM AND SENSING', and 'QUORUM AND QUENCHING', accessed on 15 February 2022

marked by full colonization, then finally develops into a fully established biofilm on the equipment surface. The film has a strong structure due to polymer-like substances called Extra-cellular Polymeric Substances (EPS), protecting it from biocides and other toxins [35].

Introduction to quorum sensing and its role in membrane biofouling: an overview

Cell-to-cell communication occurring between microbes is termed as quorum sensing (QS) [36], and it has close association with biofouling. This communication between microbial cells was first observed in 1965 by Tomasz [37], where 'hormone-

like activators' showed a significant part in bacterial propagation. These activators are now termed as autoinducers and serve as QS signals in bacteria. Microbes make use of several QS signals [38] for the same, such as autoinducer peptides (AIPs), N-acyl homoserine lactones (AHLs) [39], and autoinducer-2 (AI-2). Other molecule identifiers are 3-hydroxy palmitic acid methyl ester [40] and volatile diffusible signal factor [41]. Figure 3 illustrates probable QQ strategies for control of biofouling [42]

Figure 4 gives a view of the recent interest in the above in the past decade, as well as in comparison to quorum quenching:

As illustrated above, there has been a growing interest in QS studies across various fields, such

Table 2. QS signals generated by different bacteria, along with the behaviour that these signals help the bacteria exhibit.

QS bacteria	Signal molecule	Behaviour	Reference
<i>Streptococcus intermedius</i>	CSP	Biofilm formation	[47]
<i>Streptococcus pneumoniae</i>		Genetic competence	[37]
<i>Vibrio fischeri</i>	LuxI and LuxR	Luminescence	[48]
<i>V. fischeri</i>	N-3-oxohexanoyl-L-homoserine lactone	Biofilm formation	[49]
<i>Vibrio cholerae</i>	hapR	Biofilm formation	[50]
α -, β - and γ -Proteobacteria	N-acyl-homoserine lactones	Aerobic granulation, Biofilm formation, and EPS production	[51]
<i>Vibrio harveyi</i>	Autoinducer 2 (AI-2)	Swarming motility and swarming motility	[52]
Gram-positive bacteria	Autoinducing peptides (AIP)	Intercellular communication	[53]
Gastrointestinal microbial flora	Autoinducer 3 (AI-3)	Intercellular communication	[54]
<i>Pseudomonas</i>	<i>Pseudomonas</i> quinolone signal (PQS)	Biofilm formation and transport regulation	[55]
<i>Ralstonia solanacearum</i>	3-Hydroxypalmitic acid methyl ester	AHL production	[56]

as synthetic biology [43], food industry [44], microbiology [45], anti-virulence therapy [46] and more. This has resulted in more information regarding QS bacteria and their signals, using various characterization techniques for the same. Table 2 below gives a brief overview of some major QS bacteria.

These studies have provided for a decent understanding of the various means through which QS occurs in bacteria. For example, biofilm formation due to bacteria on the membrane surface is due to the involvement of QS. Studies conducted on several bacteria like *Burkholderia thailandensis*, *Rhodobacter sphaeroides*, *Pantoea ananatis*, and *E. coli* prove the same [57]. Shrout and Nerenberg [58], in their review reported that these biofilm formations occur after the concentration of the signalling molecules reaches a certain threshold, above which there is mutual growth prompted along with production of EPS. The formation of aerobic sludge granules [51], stabilization of microbial communities and exoenzyme production [59] also requires QS signal manipulation.

QS follows three steps [60]: (i) the microbial cell community produces small diffusible chemical signals, (ii) receptors in the cell detect these chemicals when the concentration exceed a threshold, and (iii) activation of specific target genes by detection of autoinducers while allowing for more production of autoinducers.

In the bioformation stage of biofouling, QS plays a major role. Biofilm formation may be defined by four stages: (i) single-cell attachment onto the membrane surface, (ii) EPS production, providing surface adhesion and promotion biofilm structure development, (iii) biofilm development enabled by aggregation of cells, and (iv) single cell dispersion from the biofilm [61].

Hence, several studies have been conducted to understand the role of QS in biofilm formation, but for the first record, it was Davies et al. [62] in 1998 who reported a possible connection between biofilm formation and quorum sensing, finding that the lack of AHL synthase gene, *lasI*, in *P. aeruginosa* PAO1. They concluded that this lack of the gene was the cause of the homogenous and flat biofilm formed that was largely different from the heterogeneous and highly structured biofilm formed by wild type PAO1. Subsequently, more research has been done on the effect of QS on biofilm formation by

others. This has mainly been for bacterial species such as *Serratia liquefaciens*, *Burkholderia cepacia* H111, *Aeromonas hydrophila*, and *Streptococcus* mutants [50,63].

Thus, an understanding and inhibition of this communication would prove beneficial to stop biofouling, and after a lot of study, in 2009, Yeon et al. [64] proved quorum quenching to be a promising method for the same. In their study, a QQ enzyme was immobilized on magnetic particles to prepare a magnetic enzyme carrier (MEC). They then studied the activity trend in the MEC for both short term and long term and found that there was no reduction in under continuous shaking for both iterative cycles (14 and 29 days respectively) of reuse. In comparison, a free enzyme in a batch type MBR is shown to have less efficiency in combating biofouling, recyclability, and in mixed liquor. When operated in lab scale, the MEC showed an enhanced membrane permeability to a higher degree as compared to a MBR with no enzyme. Thus, quorum quenching was proving to be an attractive method of countering biofouling, while maintaining the survivability and growth of the microbial community [65].

Quorum quenching: an overview

The process of disrupting, stopping, or regulating microbial communication is termed as quorum quenching (QQ). This degradation of signals involves biotic and abiotic factors, such as pH, enzymatic actions, environmental factors, etc. [66]. There are several benefits to using quorum quenching [67]: Show lesser selective pressure as compared to antibiotics, shows the least amount of impact on host

Table 3. QS signals generated by different bacteria, along with the behaviour that these signals help the bacteria exhibit.

QQ microorganism	QQ signal	Reference
<i>A. bereziniae</i> , <i>A. junii</i>	C6-HSL, 3-oxo-C12-HSL	[68]
<i>Bacillus firmus</i> , <i>B. megaterium</i>	C10-HSL, 3-oxo-C12-HSL	
<i>Bosea thioxidans</i>	3-oxo-C6,8,12-HSL	[69]
<i>M. luteus</i>	C6-HSL	
<i>A. delafieldii</i>	3-oxo-C12-HSL	[70]
<i>Chryseobacterium indologenes</i>	C10-HSL, 3-oxo-C12-HSL	
<i>Diaphorobacter nitroreducens</i>	3-oxo-C12-HSL	[71]
<i>Enterobacter ludwigii</i>	3-oxo-C6-HSL	[61]
<i>Mesorhizobium ciceri</i>	3-oxo-C12-HSL	[72]
<i>Ochrobactrum anthropi</i>	3-oxo-C6,8,12-HSL	[73]
<i>Rheinheimera chironomi</i>	3-oxo-C12-HSL	[63]
<i>Roseomonas terrae</i>	3-oxo-C6, 8, 12-HSL	[74]
<i>Trichosporon montevidense</i>	3-oxo-C12-HSL	[75]
<i>Lactobacillus</i> sp. SBR04MA	C6-HSL	[76]

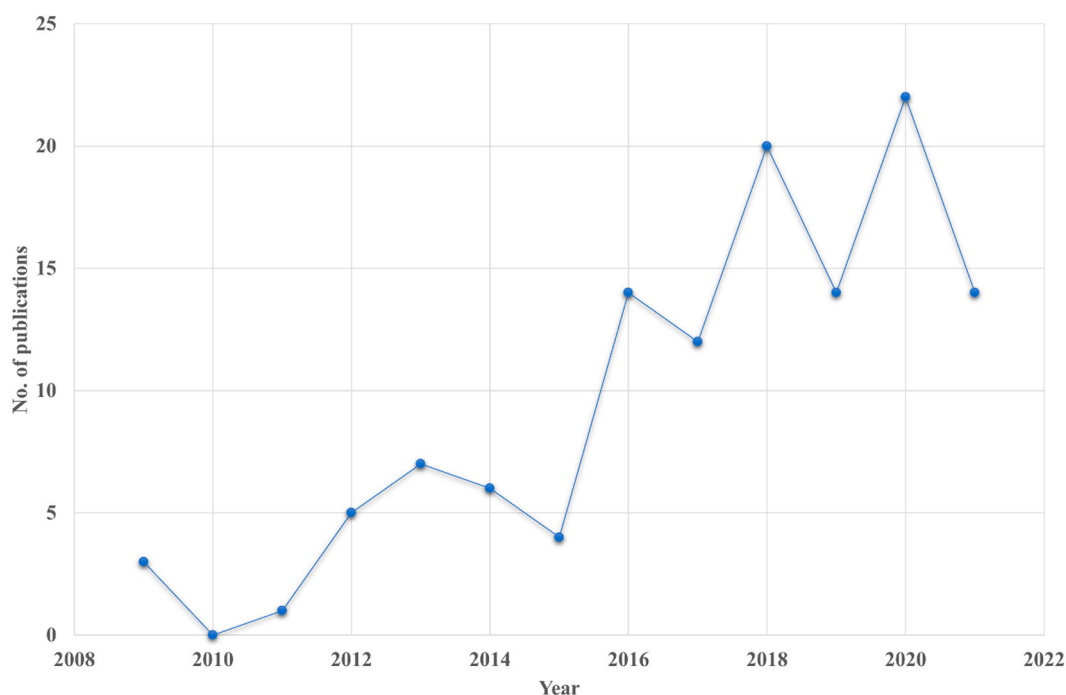


Figure 5. Number of publications on quorum quenching in biofouling versus the year of publication in the past decade. Data taken from SCOPUS with keywords 'BIOFOULING AND QUORUM AND QUENCHING', accessed on 15 February 2022

flora, rapid inactivation of the microorganisms who are to be targeted, increased efficacy of antibiotics due to supplementation, enables better protection for immunocompromised individuals, enables the blocking multiple virulence factors by stopping them from producing secretions.

Some of the major QQ strain producing microorganisms are listed below in Table 3:

Three strategies have been developed as possible ways to inhibit a QS system:

- (1) blockage of the microorganism's signal synthesis,
- (2) inactivation of the microorganism's signals, and
- (3) interfering with cell-to-cell communication by disrupting the signal receptor [77]. Out of these, inactivation of signal is the most effective means of QQ as it has the least probability to affect the cellular functions of the organism. Kim et al. [78] managed to mitigate the formation of fungal biofilm in a nanofiltration process by making use of QQ enzyme immobilized on the membrane. The biofilm was reduced to less production of EPS, which was due to the immobilized membrane maintaining more than 90% of its activity for several cycles of the reaction and washing. Uroz et al. [79] also succeeded in activating

AHLs by using *Rhodococcus erythropolis* strain W2 with oxidoreductase and amidolytic reactions.

Membrane biofouling control using quorum quenching (QQ)

Biofilm formation in membranes has a slightly different mechanism, particularly in membrane bioreactors. As mentioned previously with respecting to biofouling, membrane biofouling is caused through the following steps:

- (1) microbial deposition of cells and products onto the membrane surface [80],
- (2) growth of the microbial cells on the membrane surface by their multiplication [81], and
- (3) production of EPS matrix to encase the cells for film formation [82].

Owing to the above, quorum quenching has been developed as a novel tool for anti-fouling techniques, particularly in the case of bioreactors. However, given that it is still in development, and particularly with respect to biofouling, there have not been many studies conducted, as seen in Figure 5. Further, there is a need to understand the effective dosages of the QQ bacteria required, as well as involvement of

different times of bacteria since at present, N-acyl homoserine lactones (AHLs) are most frequently used [83], and the feasibility of these methods in an anerobic state has not been widely studied. An efficient biofouling control method developed is the enzymatic quenching of AHL signals using lactonases or acylases. Many bacteria are known to produce these enzymatic quenching mechanisms and have been seen in bacteria that produce enzymes [84], such as *Agrobacterium tumefaciens* C58, *P. aeruginosa* PAO1, *Rhodococcus erythropolis* strain W2, *Anabaena* sp. PCC7120, *Ralstonia* sp. XJ12B, *Bacillus* sp. strain 240B1, etc. Interspecies QQ by bacteria has also been seen as much more efficient and economic than otherwise, owing to high extraction and purification cost due to low stability of enzymes within the same species.

Quorum quenching in membrane reactors

There have been multiple attempts in using the QQ action of bacterial cells to mitigate biofouling in membrane bioreactors. Kim et al. [85] made use of QQ bacteria entrapping free-moving beads to inhibit biofouling in a MBR. The QQ bacteria used by them was *Rhodococcus* sp. BH4, which were entrapped in alginate beads to form cell entrapping beads. This forced the microbial cells on the film to produce low amounts of extracellular polymeric substances, enabling a loose biofilm formation which could be easily removed by physical detachment. In another study by Oh et al. [85] using the same QQ bacteria *Rhodococcus* sp. BH4 along with *E. coli* encapsulated inside the lumen of the hollow fibre membrane, it was found that even across 80 days of use, the biofouling control was successful.

Mukerji et al. [86] made use of *Kluyvera citrophila* penicillin G acylase as a novel molecule to cleave the QS molecules (AHLs), owing to its increased

resilience and ease of immobilization. They discovered that the AHLs would fit perfectly into the hydrophobic pocket of the activation site of the enzyme, making it very efficient as a QQ molecule.

Reactor design for the purpose of QQ have also been employed in aerobic membrane reactors. Ergon-Can et al. [87] demonstrated a novel bacterial immobilization medium design for the inhibition of biofilm formation in an MBR by using microfiltration membranes on polycarbonate frames entrapped with *Rhodococcus* sp. BH4. They observed the QQ effect in these reactors both long term and short term and were able to discern efficiency over a long term. They also discovered that the design effected the microbial diversity as well, by preventing some common species from dominating over others in the microbial community. Similarly, Lee et al. [88] stabilized and immobilized acylase in magnetically separable mesoporous silica and maintained its anti-fouling activity. This was highly sustainable and efficient even under harsh conditions of high organic loading and low dosage of enzyme.

While the above has been for aerobic membrane bioreactors and show lot of promise, most of the methods that have presently been explored for the mitigation of biofouling in anaerobic membrane reactors (AnMBR) show a reduced efficiency [89]. Further, in comparison to an aerobic MBR, the application of QQ for the mitigation of biofouling on AnMBR is still unclear [90]. For the working of QQ under anaerobic conditions, anaerobic or facultative bacteria that are responsible for QQ should be selected. In a study by Kim et al. [91], were 225 QQ bacteria were isolated, it was determined (and later confirmed by Wang et al. [92]) that the QQ functions of *Microbacterium* was potentially applicable in anerobic conditions as well. Based off this, Liu et al. [89] made use of *Microbacterium* sp embedded in alginate beads to mitigate biofouling in an AnMBR. The AnMBR showed an increased lifespan of about 8–10 times more during its operation period, proving it to be a useful method.

In general, QQ_MBRs show much better biofouling resistance and management compared to conventional MBRs. However, QQ-MBRs are still far from replacing other anti-biofouling techniques entirely, or from removing the need for mitigation procedures through physical and chemical cleaning.

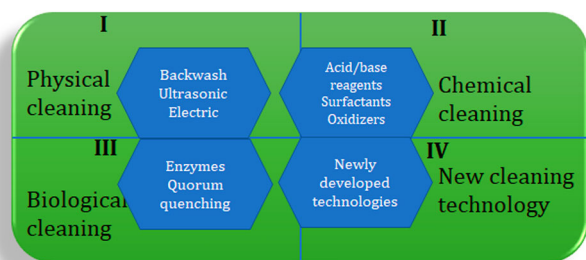


Figure 6. Various methods for membrane cleaning and fouling mitigation.

However, this also means that an effective method to mitigate biofouling is to combine QQ with the above cleaning methods. Weerasekara et al. [93] studied the combination of chemically enhanced physical cleaning in a QQ-vessel and reported that the combination of chemically enhanced back washing in a bacterial QQ-vessel caused effective mitigation of biofouling. QQ enabled in easy physical removal of the attached biofilm, while the chemical used mitigated the chemically reversible filtration resistance.

Media developed to enable QQ in MBR

Owing to these benefits of QQ on MBRs, multiple devices have been developed to enable them. Some features of such devices are [94]: Protection of QQ bacteria against other microorganisms, Promotion of QQ bacterial growth, Showing increased stability and resistance to adverse conditions, Allowing access to filtration membrane for easy monitoring and study.

QQ vessel

In the earlier mentioned study conducted by Oh et al. [95] the lumen of a microporous hollow fibre membrane was used to encapsulate QQ bacteria. This formed the first QQ vessel and proved to be an effective anti-biofouling media, by showing anti-biofouling effect maintained in an operation of over 100 days. In another study, Jahangir et al. [96] noticed that a QQ-vessel containing BH4 was also highly effective, particularly when the QQ bacteria had more access to AHLs, when it was closer to the filtration membrane.

However, in long term, there was reduction in MBR due to the QQ bacteria present in high density. This was mainly due to reduced food for the QQ bacteria. To combat this, a novel QQ-vessel design [97] was proposed. Here, the MBR feed was directly fed into the lumen of the QQ-vessel, enabling it to retain more bacterial QQ activity in the inner flow mode. This helped in better mitigation of biofouling as compared to the standard mode. A rotating QQ-vessel was also proposed as an alternative, where a polycarbonate rotating microbial carrier frame (RMCF) lined with flat PVDF microfiltration membrane sheets was used. This had a greater biofouling mitigation than a regular QQ-vessel. Cheong et al. [98] also attempted

to combat the problem of high density of QQ bacteria using a monolithic ceramic microporous membrane with seven lumens.

QQ-bead

Another media to entrap QQ bacteria that was proposed were beads. This was made possible by entrapping QQ bacteria on beads made of alginate or polyvinyl alcohol beads through their microporous structures. These QQ beads could circulate freely within the MBR, unlike the QQ-vessel. This increased the otherwise low AHL degradation effectiveness, as QQ-beads can freely circulate in the bulk liquid and have larger contact with the biofilm on the filtration membrane surface, enabling the trapping of AHLs more efficiently. Kim et al. [85] found that the combined effect of QQ and physical washing was 10 times more effective when QQ beads were used, as compared to a conventional MBR. They made use of *Bacillus methylophilus* sp. WY entrapped in alginate beads to form the QQ beads and showed boosted membrane efficiency that was increased by 3–4 times. For a wide spectrum of AHLs, the strain WY showed better degradation efficiency than *Pseudomonas* sp. 1A1 and *Rhodococcus* sp. BH4. Xiao et al. [99] used a combination of QQ and powdered activated carbon in a lab-scale MBR, with QQ-PAC-alginate beads, and showed easy removal of chemicals from the feed.

QQ-cylinder and QQ-hollow cylinder

While QQ-beads showed high efficiency, their surface area of contact at a time leaved room for more. The idea of a cylinder and hollow cylinder was thus proposed to maximize the surface area of the cell entrapping medium [100]. These cylinder designs showed more promise than QQ-beads, especially after the cylinder's surface area was increased. The cylindrical shape in general was proven to be more effective than the spherical shape, rendering these more effective than the QQ-beads. As a result, physical washing became more efficient.

QQ-sheet

While the above media are used in MBRs with flat membranes, they face difficulty in use in MBRs with dense structures like fibre bundles. Nahm et al. [101] developed a QQ bacteria entrapping sheet (QQ-sheet) to solve this, where QQ bacteria were entrapped in sheets. These showed 2.5 times more

activity than other QQ media, owing to their larger surface area.

Fouling mitigation strategies

Depending on foulants composition, membrane materials, and the nature of reagents, membrane cleaning primarily includes chemical, physical and biological. Various methods are illustrated in [Figure 6](#). Physical cleaning is not able to treat irreversible fouling, and thus membrane efficiency (flux) could be restored merely by chemical cleaning. While chemical cleaning has high efficiency and could be applied for irreversible fouling, strong acids and alkalis damage membrane [\[102\]](#).

A coating possessing superamphiphobic characteristics (both superoleophobic and superhydrophobic) was developed using needle-shaped chitin nanocrystals that had been altered with thiol groups and elongated fluorinated chains. This innovative coating exhibited effective self-cleaning and antifouling attributes when subjected to an oil solution. Although superoleophobic surfaces are effective towards natural compounds, they aren't appropriate for filtering oil and water due to their intense hydrophobicity, lacking the ability to differentiate between oil and water [\[103\]](#)

While marine antifouling substances offer a straightforward solution to biofouling issues compared to traditional coatings that result in secondary pollutants, their sturdiness in marine environments remains complicated, often resulting in coating wear and detachment from substrates [\[15\]](#).

Exploiting various enzymes have demonstrated effectiveness in controlling and eliminating biofouling also being eco friendly to membranes and surroundings. This approach stems from an understanding of biofouling's nature. Enzymes can break down quorum sensing biomolecules that regulate microbial communication for biofilm formation, as well as the extracellular polymeric substances matrix that supports microbial growth. The primary purpose of using enzymes is to hydrolyse these biopolymers, thereby managing biofouling [\[31\]](#). Chemically active antifouling coatings, which include biocide-based and enzyme-primarily based alternatives, have proven effective but pose risks to non-target marine existence because of harmful components like organotin [\[104\]](#). Strategies

involving nanoparticles in membrane composition have improved our understanding however can cause structural troubles like terrible particle-polymer interactions and choppy distribution, complicating the composite membrane's residences [\[105\]](#).

Self-polishing copolymer paints, freeing toxic tributyltin throughout vessel operation, have dangerous outcomes on aquatic species. TBT-integrated antifouling paints cause numerous unfavourable effects, inclusive of enforcing male traits on female species at low concentrations [\[106\]](#). Diverse strategies have emerged to counter biofilm formation on medical devices: incorporating biocidal dealers, changing polymer homes for antifouling surfaces, or combining antifouling and bactericidal developments. Yet, prolonged use of biocidal coatings has been linked to rising antibiotic resistance. While biomimetic surfaces offer a green strategy to biofouling, they continue to be unavailable commercially and sourcing sufficient compounds presents demanding situations [\[107\]](#).

Conclusions

Quorum sensing techniques have paved way to quorum quenching methods as a novel method for anti-biofouling techniques, particularly in the case of MBRs. While there are several studies that have been conducted on this and may more to come, there is still a dearth of studies done to make these QQ techniques more economical and accessible. Particularly for commercial use at a large scale, there still needs to be better research on the optimal conditions, design of QQ media, and practical use of QQ media. Further, while the goal of research on this topic is to mitigate biofouling, a lot of these studies lean towards microbiology and MBRs alone, ignoring other areas such as waste treatment, industrial fouling, etc. With more diverse research done on QQ, it will be beneficial to push QQ onto other industries and equipment as well, particularly those of membrane technology beyond MBRs. While the potential of these have been demonstrated in a lab scale, the commercial usage remains to be seen.

Membrane fouling is a serious issue in many industries. The fouling mitigation approaches currently applied accounts for a large percentage of total energy requirements. Chief economic and environmental considerations are important for

comprehending the potential effects of executing anti-fouling strategies. Numerous factors influencing microbial deposition include membrane properties, hydrodynamic conditions, feed solution compositions, and membrane module designs. Biofouling mitigation is crucial as it impacts membrane longevity, raises operational costs, and compromises the quality of permeate water. Quorum quenching is a promising technique against bacteria, yet there still remains a lot to be done, before it can be fully integrated into various fields of study. These biological methods are found to be successful in mitigating fouling in membrane reactors. In spite of the recent advancements, many prevailing challenges limit the integrated and large scale applications of this technology.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

No data was used for the research described in the article.

Notes on contributors

Anusha Yethadka Ganaraja is a recent undergrad in Chemical Engineering from Manipal Institute of Technology, India. Her research interests and past projects include adsorption, wastewater treatment, and design of experiments. She is currently pursuing her Master of Science in Sustainable energy Technology in Delft University of Technology, Netherland.

Dr Lavanya Mulky is an Assistant professor in Department of Chemical Engineering, Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, Karnataka - India. She has received her doctoral degree from Manipal Academy of Higher Education, Manipal, Karnataka - India. She has published 35 research papers in peer-reviewed journals. Her scientific focus lies in the areas of Erosion-corrosion, Microbial corrosion in industrial metals, hydrodynamics of biofilms, bioremediation.

References

- [1] Anis SF, Hashaikeh R, Hilal N. Microfiltration membrane processes: A review of research trends over the past decade. *J Water Process Eng.* 2019;32:1–12. doi:10.1016/j.jwpe.2019.100941.
- [2] Al Aani S, Mustafa TN, Hilal N. Ultrafiltration membranes for wastewater and water process engineering: A comprehensive statistical review over the past decade. *J Water Process Eng.* 2020;2020(35):1–14. doi:10.1016/j.jwpe.2020.101241.
- [3] Mohammad AW, Teow YH, Ang WL, et al. Nanofiltration membranes review: recent advances and future prospects. *Desalination.* 2015;356:226–254. doi:10.1016/j.desal.2014.10.043
- [4] Hailemariam RH, Woo YC, Damtie MM, et al. Reverse osmosis membrane fabrication and modification technologies and future trends: A review. *Adv Colloid Interface Sci.* 2020;276:102100.
- [5] Awad AM, Jalab R, Minier-Matar J, et al. The status of forward osmosis technology implementation. *Desalination.* 2019;461:10–21. doi:10.1016/j.desal.2019.03.013
- [6] Weerasekara NA, Woo SG, Criddle C, et al. Clues to membrane fouling hidden within the microbial communities of membrane bioreactors. *Environ Sci Water Res Technol.* 2019;5:1389–1399. doi:10.1039/C9EW00213H
- [7] Shahid MK, Pyo M, Choi YG. Inorganic fouling control in reverse osmosis wastewater reclamation by purging carbon dioxide. *Environ Sci Pollut Res.* 2019;26:1094–1102. doi:10.1007/s11356-017-9008-3
- [8] Mansi AE, El-Marsafy SM, Elhenawy Y, et al. Assessing the potential and limitations of membrane-based technologies for the treatment of oilfield produced water. *Alexandria Eng J.* 2022;68:787–815. doi:10.1016/j.aej.2022.12.013
- [9] Gkotsis PK, Mitrakas MM, Tolkou AK, et al. Batch and continuous dosing of conventional and composite coagulation agents for fouling control in a pilot-scale MBR. *Chem Eng J.* 2017;311:255–264. doi:10.1016/j.cej.2016.11.099
- [10] Gkotsis PK, Banti DC, Peleka EN, et al. Fouling issues in membrane bioreactors (MBRs) for wastewater treatment: major mechanisms, prevention and control strategies. *Processes.* 2014;2(4):795–866. doi:10.3390/pr2040795
- [11] Nguyen T, Roddick FA, Fan L. Biofouling of water treatment membranes: a review of the underlying causes, monitoring techniques and control measures. *Membranes.* 2014;2(4):804–840. doi:10.3390/membranes2040804
- [12] Cui Y, Gao H, Yu R, et al. Biological-based control strategies for MBR membrane biofouling: A review. *Water Sci Tech.* 2021;83(11):2597–2614. doi:10.2166/wst.2021.168
- [13] Nebot E, Casanueva JF, Casanueva T, et al. Model for fouling deposition on power plant steam condensers cooled with seawater: effect of water velocity and tube material. *Int J Heat Mass Transf.* 2007;17–18:3351–3358.
- [14] Loxton J, Macleod AK, Nall CR, et al. Setting an agenda for biofouling research for the marine renewable energy industry. *Int J Mar Energy.* 2017;19:292–303. doi:10.1016/j.ijome.2017.08.006
- [15] Wang F, Zhang H, Yu B, et al. Review of the research on anti-protein fouling coatings materials. *Prog Org Coat.* 2020;147:1–16. doi:10.1016/j.porgcoat.2020.105860

- [16] Abd El Aleem FA, Al-Sugair KA, Alahmad MI. Biofouling problems in membrane processes for water desalination and reuse in Saudi Arabia. *Int Biodeter Biodegrad.* **1998**;41(1):19–23. doi:[10.1016/S0964-8305\(98\)80004-8](https://doi.org/10.1016/S0964-8305(98)80004-8)
- [17] Omae I. Organotin antifouling paints and their alternatives. *Applied Organ Chem.* **2003**;17(2):81–105. doi:[10.1002/aoc.396](https://doi.org/10.1002/aoc.396)
- [18] Drews A. Membrane fouling in membrane bioreactors—characterisation, contradictions, cause and cures. *J Memb Sci.* **2010**;363(1–2):1–28. doi:[10.1016/j.memsci.2010.06.046](https://doi.org/10.1016/j.memsci.2010.06.046)
- [19] Iqbal T, Shah SSA, Lee K, et al. Porous shell quorum quenching balls for enhanced anti-biofouling efficacy and media durability in membrane bioreactors. *Chem Eng J.* **2021**;406:126869.
- [20] Vinardell S, Cortina JL, Valderrama C. Environmental and economic evaluation of implementing membrane technologies and struvite crystallisation to recover nutrients from anaerobic digestion supernatant. *Biores Technol.* **2023**;1–11.
- [21] Bhattacharyya A, Liu L, Walsh M, et al. Membrane technology for treating decanted oily wastewater from marine oil spill operations: comparison between membrane filtration and membrane bioreactor. *Mar Pollut Bull.* **2023**;194:1–15. doi:[10.1016/j.marpolbul.2023.115397](https://doi.org/10.1016/j.marpolbul.2023.115397).
- [22] Di Bella G, Di Prima N, Di Trapani D, et al. Performance of membrane bioreactor (MBR) systems for the treatment of shipboard slops: assessment of hydrocarbon biodegradation and biomass activity under salinity variation. *J Hazard Mater.* **2015**;300:765–778. doi:[10.1016/j.jhazmat.2015.08.021](https://doi.org/10.1016/j.jhazmat.2015.08.021)
- [23] Nasir AM, Adam MR, Kamal SNEAM, et al. A review of the potential of conventional and advanced membrane technology in the removal of pathogens from wastewater. *Sep Purif Technol.* **2022**;286:120454. doi:[10.1016/j.seppur.2022.120454](https://doi.org/10.1016/j.seppur.2022.120454)
- [24] Zhao Z, Muylaert K, Vankelecom IF. Applying membrane technology in microalgae industry: A comprehensive review. *Renew Sust Energy Rev.* **2023**;172:113041. doi:[10.1016/j.rser.2022.113041](https://doi.org/10.1016/j.rser.2022.113041)
- [25] Pasichnyk M, Stanovsky P, Polezhaev P, et al. Membrane technology for challenging separations: removal of CO₂, SO₂ and NO_x from flue and waste gases. *Sep Purif Technol.* **2023**; 124436. doi:[10.1016/j.seppur.2023.124436](https://doi.org/10.1016/j.seppur.2023.124436)
- [26] Gkotsis P, Kougias P, Mitrakas M, et al. Biogas upgrading technologies—recent advances in membrane-based processes. *Int JHydr Energy.* **2023**;48(10):3965–3993. doi:[10.1016/j.ijhydene.2022.10.228](https://doi.org/10.1016/j.ijhydene.2022.10.228)
- [27] Jafari M, Vanoppen M, van Agtmaal JMC, et al. Cost of fouling in full-scale reverse osmosis and nanofiltration installations in The Netherlands. *Desalination* **2021**; 500:1–11. doi:[10.1016/j.desal.2020.114865](https://doi.org/10.1016/j.desal.2020.114865).
- [28] Archana S, Sundaramoorthy B, Faizullah M. Review on impact of biofouling in aquafarm infrastructures. *Int J Curr Microbiol App Sci.* **2019**;8(1):2942–2953. doi:[10.20546/ijcmas.2019.807.365](https://doi.org/10.20546/ijcmas.2019.807.365)
- [29] Wittmann E, Belkhdja M, Lee Y, et al. RO membrane biofouling: problems and solutions. In *World Congress Perth Convention and Exhibition Centre 2011*; 4–9.
- [30] Jadhav DA, Pandit S, Sonawane JM, et al. Effect of membrane biofouling on the performance of microbial electrochemical cells and mitigation strategies. *Bioresour Technol Rep.* **2021**;15:1–15. doi:[10.1016/j.biteb.2021.100822](https://doi.org/10.1016/j.biteb.2021.100822).
- [31] Bachosz K, Vu MT, Nghiem LD, et al. Enzyme-based control of membrane biofouling for water and wastewater purification: A comprehensive review. *Env Tech Innov.* **2022**;25:1–16. doi:[10.1016/j.eti.2021.102106](https://doi.org/10.1016/j.eti.2021.102106).
- [32] Mansouri J, Harrisson S, Chen V. Strategies for controlling biofouling in membrane filtration systems: challenges and opportunities. *J Mater Chem.* **2010**;20(22):4567–4586. doi:[10.1039/b926440j](https://doi.org/10.1039/b926440j)
- [33] Liu C, Zhu L, Chen L. Effect of salt and metal accumulation on performance of membrane distillation system and microbial community succession in membrane biofilms. *Water Res.* **2020**;177:115805. doi:[10.1016/j.watres.2020.115805](https://doi.org/10.1016/j.watres.2020.115805)
- [34] Harris L, Tozzi S, Wiley P, et al. Potential impact of biofouling on the photobioreactors of the Offshore Membrane Enclosures for Growing Algae (OMEGA) system. *Bioresour Technol.* **2003**;144:420–428. doi:[10.1016/J.BIORTECH.2013.06.125](https://doi.org/10.1016/J.BIORTECH.2013.06.125)
- [35] Al-Juboori RA, Yusaf T. Biofouling in RO system: mechanisms, monitoring and controlling. *Desalination.* **2012**;302:1–23. doi:[10.1016/j.desal.2012.06.016](https://doi.org/10.1016/j.desal.2012.06.016)
- [36] Waters CM, Bassler BL. Quorum sensing: cell-to-cell communication in bacteria. *Annu Rev Cell Dev Biol.* **2005**;21:319–346. doi:[10.1146/annurev.cellbio.21.012704.131001](https://doi.org/10.1146/annurev.cellbio.21.012704.131001)
- [37] Tomasz A. Control of the competent state in pneumococcus by a hormone-like cell product: an example for a New type of regulatory mechanism in bacteria. *Nature.* **1965**;208(5006):155–159. doi:[10.1038/208155a0](https://doi.org/10.1038/208155a0)
- [38] Neves PR, McCulloch JA, Mamizuka EM, et al. PSEUDOMONAS | *Pseudomonas aeruginosa*. In: *Encyclopedia of food microbiology: second edition*. Academic press; **2014**. p. 253–260.
- [39] Ampomah-Wireko M, Luo C, Cao Y, et al. Chemical probe of AHL modulators on quorum sensing in gram-negative bacteria and as antiproliferative agents: A review. *Eur J Med Chem.* **2021**;226:1–25. doi:[10.1016/j.ejmech.2021.113864](https://doi.org/10.1016/j.ejmech.2021.113864).
- [40] Weisskopf L, Schulz S, Garbeva P. Microbial volatile organic compounds in intra-kingdom and inter-kingdom interactions. *Nat Rev Microbiol.* **2021**;19(6):391–404. doi:[10.1038/s41579-020-00508-1](https://doi.org/10.1038/s41579-020-00508-1)
- [41] Zhou L, Zhang LH, Cámara M, et al. The DSF family of quorum sensing signals: diversity, biosynthesis, and turnover. *Trends Microbiol.* **2017**;25(4):293–303. doi:[10.1016/j.tim.2016.11.013](https://doi.org/10.1016/j.tim.2016.11.013)

- [42] Lee K, Yu H, Zhang X, et al. Quorum sensing and quenching in membrane bioreactors: opportunities and challenges for biofouling control. *Biores Tech.* **2018**;270:656–668. doi:10.1016/j.biortech.2018.09.019
- [43] Boo A, Ledesma Amaro R, Stan GB. Quorum sensing in synthetic biology: A review. *Curr Opin Syst Biol.* **2021**;28:100378. doi:10.1016/j.coisb.2021.100378
- [44] Machado I, Silva LR, Giaouris ED, et al. Quorum sensing in food spoilage and natural-based strategies for its inhibition. *Food Res Int.* **2020**;127:1–12. doi:10.1016/j.foodres.2019.108754.
- [45] Sharma A, Singh P, Sarmah BK, et al. Quorum sensing: its role in microbial social networking. *Res Microbiol.* **2020**;171(5–6):159–164. doi:10.1016/j.resmic.2020.06.003
- [46] Defoirdt T. Quorum-sensing systems as targets for anti-virulence therapy. *Trends Microbiol.* **2018**;26(4):313–328. doi:10.1016/j.tim.2017.10.005
- [47] Petersen FC, Pecharki D, Scheie AA. Biofilm mode of growth of *Streptococcus intermedius* favored by a competence-stimulating signaling peptide. *J Bacteriol.* **2004**;186(18):6327–6331. doi:10.1128/JB.186.18.6327-6331.2004
- [48] Engebrecht J, Nealson K, Silverman M. Bacterial bioluminescence: isolation and genetic analysis of functions from *Vibrio fischeri*. *Cell.* **1983**;32(3):773–781. doi:10.1016/0092-8674(83)90063-6
- [49] Engebrecht JA, Silverman M. Identification of genes and gene products necessary for bacterial bioluminescence. *Proc Natl Acad Sci USA.* **1984**;81(13):4154–4158. doi:10.1073/pnas.81.13.4154
- [50] Hammer BK, Bassler BL. Quorum sensing controls biofilm formation in *Vibrio cholerae*. *Mol Microbiol.* **2003**;50(1):101–104. doi:10.1046/j.1365-2958.2003.03688.x
- [51] Tan CH, Koh KS, Xie C, et al. The role of quorum sensing signalling in EPS production and the assembly of a sludge community into aerobic granules. *ISME J.* **2014**;8(6):1186–1197. doi:10.1038/ismej.2013.240
- [52] Tarighi S, Taheri P. Different aspects of bacterial communication signals. *World J Microbiol Biotech.* **2010**;27(6):1267–1280. doi:10.1007/s11274-010-0575-4
- [53] Sturme MHJ, Kleerebezem M, Nakayama J, et al. Cell to cell communication by autoinducing peptides in gram-positive bacteria. *Antonie Van Leeuwenhoek.* **2002**;81(1–4):233–243. doi:10.1023/A:1020522919555
- [54] Clarke MB, Hughes DT, Zhu C, et al. The QseC sensor kinase: A bacterial adrenergic receptor. *PNAS.* **2006**;103(27):10420–10425. doi:10.1073/pnas.0604343103
- [55] Diggle SP, Cornelis P, Williams P, et al. 4-Quinolone signalling in *Pseudomonas aeruginosa*: Old molecules, new perspectives. *Int J Med Microbiol.* **2006**;296(2–3):83–91. doi:10.1016/j.ijmm.2006.01.038
- [56] Flavier AB, Ganova-Raeva LM, Schell MA, et al. Hierarchical autoinduction in *Ralstonia solanacearum*: control of acyl-homoserine lactone production by a novel autoregulatory system responsive to 3-hydroxypalmitic acid methyl ester. *J Bacteriol.* **1997**;179(22):7089–7097. doi:10.1128/jb.179.22.7089-7097.1997
- [57] Whiteley M, Diggle SP, Greenberg EP. Progress in and promise of bacterial quorum sensing research. *Nature.* **2017**;551(7680):313–320. doi:10.1038/nature24624
- [58] Shrout JD, Nerenberg R. Monitoring bacterial Twitter: does quorum sensing determine the behavior of water and wastewater treatment biofilms? *Environ Sci Technol.* **2012**;46(4):1995–2005. doi:10.1021/es203933h
- [59] Chong G, Kimyon O, Rice SA, et al. The presence and role of bacterial quorum sensing in activated sludge. *Microb Biotechnol.* **2012**;5(5):621–633. doi:10.1111/j.1751-7915.2012.00348.x
- [60] Oh HS, Lee CH. Origin and evolution of quorum quenching technology for biofouling control in MBRs for wastewater treatment. *J Memb Sci.* **2018**;554:331–345. doi:10.1016/j.memsci.2018.03.019
- [61] Gopu V, Meena CK, Murali A, et al. Quorum quenching activity in the cell-free lysate of enterobacter ludwigii isolated from beef and its effect on quorum sensing regulation in *Yersinia enterocolitica*. *RSC Adv.* **2016**;6(25):21277–21284. doi:10.1039/C5RA25440J
- [62] Davies DG, Parsek MR, Pearson JP, et al. The involvement of cell-to-cell signals in the development of a bacterial biofilm. *Science.* **1998**;280(5361):295–298. doi:10.1126/science.280.5361.295
- [63] Halpern M, Senderovich Y. Chironomid microbiome. *Microb Ecol.* **2015**;70(1):1–8. doi:10.1007/s00248-014-0536-9
- [64] Yeon KM, Lee CH, Kim J. Magnetic enzyme carrier for effective biofouling control in the membrane bioreactor based on enzymatic quorum quenching. *Environ Sci Technol.* **2009**;43(19):7403–7409. doi:10.1021/es901323k
- [65] Huang J, Shi Y, Zeng G, et al. Acyl-homoserine lactone-based quorum sensing and quorum quenching hold promise to determine the performance of biological wastewater treatments: An overview. *Chemosphere.* **2016**;157:137–151. doi:10.1016/j.chemosphere.2016.05.032
- [66] Dong YH, Wang LH, Xu JL, et al. Quenching quorum-sensing-dependent bacterial infection by an N-acyl homoserine lactonase. *Nature.* **2001**;411(6839):813–817. doi:10.1038/35081101
- [67] Santhakumari S, Ravi A. Targeting quorum sensing mechanism: An alternative anti-virulent strategy for the treatment of bacterial infections. *S Afr J Bot.* **2019**;120:81–86. doi:10.1016/j.sajb.2018.09.028
- [68] Ochiai S, Morohoshi T, Kurabeishi A, et al. Production and degradation of N-acylhomoserine lactone quorum sensing signal molecules in bacteria isolated from activated sludge. *Biosci Biotechnol Biochem.* **2013**;77(12):2436–2440. doi:10.1271/bbb.130553
- [69] Tan CH, Koh KS, Xie C, et al. Community quorum sensing signalling and quenching: microbial granular biofilm assembly. *NPJ Biofilms Microbiomes.* **2015**;1(1):1–9.

- [70] Rashid R, Morohoshi T, Someya N, et al. Degradation of N-acylhomoserine lactone quorum sensing signaling molecules by potato root surface-associated *Chryseobacterium* strains. *Microbes Environ.* **2011**;26(2):1102090273. doi:[10.1264/jsme2.ME10207](https://doi.org/10.1264/jsme2.ME10207)
- [71] Chitara MK, Chauhan S, Singh RP. Bioremediation of Polluted Soil by Using Plant Growth-Promoting Rhizobacteria. **2021**(1):203–226.
- [72] Last D, Krüger GHE, Dörr M, et al. Fast, continuous, and high-throughput (bio)chemical activity assay for N-acyl-L-homoserine lactone quorum-quenching enzymes. *Appl Environ Microbiol.* **2016**;82(14):4145–4154. doi:[10.1128/AEM.00830-16](https://doi.org/10.1128/AEM.00830-16)
- [73] Krzyżanowska D, Maciąg T, Ossowicki A, et al. *Ochrobactrum quorumnecens* sp. nov., a quorum quenching bacterium from the potato rhizosphere, and comparative genome analysis with related type strains. *PLoS One.* **2022**;14(1):e0210874.
- [74] Chen JW, Gan HM, Yin WF, et al. Genome sequence of *Roseomonas* sp. Strain B5, a quorum-quenching N-acyl-homoserine lactone-degrading bacterium isolated from Malaysian tropical soil. *J Bacteriol.* **2012**;194(23):6681–6682. doi:[10.1128/JB.01866-12](https://doi.org/10.1128/JB.01866-12)
- [75] Dhagat S, Jujjavarapu SE. Isolation of a novel thermophilic bacterium capable of producing high-yield bioemulsifier and its kinetic modelling aspects along with proposed metabolic pathway. *Brazilian J Microbiol.* **2020**;51(1):135–143. doi:[10.1007/s42770-020-00228-x](https://doi.org/10.1007/s42770-020-00228-x)
- [76] Kampouris ID, Karayannakidis PD, Banti DC, et al. Evaluation of a novel quorum quenching strain for MBR biofouling mitigation. *Water Res.* **2018**;2018(143):56–65. doi:[10.1016/j.watres.2018.06.030](https://doi.org/10.1016/j.watres.2018.06.030)
- [77] Miller MB, Bassler BL. Quorum sensing in bacteria. *Ann Rev.* **2003**;55:165–199.
- [78] Kim JH, Choi DC, Yeon KM, et al. Enzyme-immobilized nanofiltration membrane to mitigate biofouling based on quorum quenching. *Environ Sci Technol.* **2011**;45(4):1601–1607. doi:[10.1021/es103483j](https://doi.org/10.1021/es103483j)
- [79] Phane Uroz S, Chhabra SR, Cá M, et al. N-Acylhomoserine lactone quorum-sensing molecules are modified and degraded by *Rhodococcus erythropolis* W2 by both amidolytic and novel oxidoreductase activities. *Microbiology.* **2005**;151:3313–3322. doi:[10.1099/mic.0.27961-0](https://doi.org/10.1099/mic.0.27961-0)
- [80] Ramesh A, Lee DJ, Wang ML, et al. Biofouling in membrane bioreactor. *Sep Sci Technol.* **2007**;41(7):1345–1370.
- [81] Lee CH, Park PK, Lee WN, et al. Correlation of biofouling with the bio-cake architecture in an MBR. *Desalination.* **2008**;231(1–3):115–123.
- [82] Krzeminski P, Leverette L, Malamis S, et al. Membrane bioreactors – A review on recent developments in energy reduction, fouling control, novel configurations, LCA and market prospects. *J Memb Sci.* **2017**;527:207–227. doi:[10.1016/j.memsci.2016.12.010](https://doi.org/10.1016/j.memsci.2016.12.010)
- [83] Müh U, Schuster M, Heim R, et al. Novel *Pseudomonas aeruginosa* quorum-sensing inhibitors identified in an ultra-high-throughput screen. *Antimicrob Agents Chemother.* **2006**;50(11):3674–3679. doi:[10.1128/AAC.00665-06](https://doi.org/10.1128/AAC.00665-06)
- [84] Lade H, Paul D, Kweon JH. N-acyl homoserine lactone-mediated quorum sensing with special reference to Use of quorum quenching bacteria in membrane biofouling control. *Biomed Res Int.* **2014**;2014:1–25.
- [85] Kim SR, Oh HS, Jo SJ, et al. Biofouling control with bead-entrapped quorum quenching bacteria in membrane bioreactors: physical and biological effects. *Environ Sci Technol.* **2013**;47(2):836–842. doi:[10.1021/es303995s](https://doi.org/10.1021/es303995s)
- [86] Mukherji R, Varshney NK, Panigrahi P, et al. A new role for penicillin acylases: degradation of acyl homoserine lactone quorum sensing signals by *Kluyvera citrophila* penicillin G acylase. *Enzyme Microb Technol.* **2014**;56:1–7. doi:[10.1016/j.enzmictec.2013.12.010](https://doi.org/10.1016/j.enzmictec.2013.12.010)
- [87] Ergön-Can T, Köse-Mutlu B, Koyuncu İ, et al. Biofouling control based on bacterial quorum quenching with a new application: rotary microbial carrier frame. *J Memb Sci.* **2017**;525:116–124. doi:[10.1016/J.MEMSCI.2016.10.036](https://doi.org/10.1016/J.MEMSCI.2016.10.036)
- [88] Lee B, Yeon KM, Shim J, et al. Effective antifouling using quorum-quenching acylase stabilized in magnetically-separable mesoporous silica. *Biomacromolecules.* **2014**;15(4):1153–1159. doi:[10.1021/bm401595q](https://doi.org/10.1021/bm401595q)
- [89] Liu J, Eng CY, Ho JS, et al. Quorum quenching in anaerobic membrane bioreactor for fouling control. *Water Res.* **2019**;156:159–167. doi:[10.1016/j.watres.2019.03.029](https://doi.org/10.1016/j.watres.2019.03.029)
- [90] Ma H, Wang X, Zhang Y, et al. The diversity, distribution and function of N-acyl-homoserine lactone (AHL) in industrial anaerobic granular sludge. *Bioresour Technol.* **2018**;247:116–124. doi:[10.1016/j.biortech.2017.09.043](https://doi.org/10.1016/j.biortech.2017.09.043)
- [91] Kim AL, Park SY, Ho LC, et al. Quorum quenching bacteria isolated from the sludge of a wastewater treatment plant and their application for controlling biofilm formation. *J Microbiol Biotechnol.* **2014**;24(11):1574–1582. doi:[10.4014/jmb.1407.07009](https://doi.org/10.4014/jmb.1407.07009)
- [92] Wang W-Z, Morohoshi T, Someya N, et al. Diversity and distribution of N-acylhomoserine lactone (AHL)-degrading activity and AHL-lactonase (AiiM) in genus microbacterium. *Microbes Environ.* **2012**;27(3):330–333. doi:[10.1264/jsme2.ME11341](https://doi.org/10.1264/jsme2.ME11341)
- [93] Weerasekara NA, Choo KH, Lee CH. Hybridization of physical cleaning and quorum quenching to minimize membrane biofouling and energy consumption in a membrane bioreactor. *Water Res.* **2014**;67:1–10. doi:[10.1016/j.watres.2014.08.049](https://doi.org/10.1016/j.watres.2014.08.049)
- [94] Bouayed N, Dietrich N, Lafforgue C, et al. Membranes process-oriented review of bacterial quorum quenching for membrane biofouling mitigation in membrane bioreactors (MBRs). *Membranes.* **2016**;6(4):52. doi:[10.3390/membranes6040052](https://doi.org/10.3390/membranes6040052)
- [95] Oh HS, Yeon KM, Yang CS, et al. Control of membrane biofouling in MBR for wastewater treatment by

- quorum quenching bacteria encapsulated in micro-porous membrane. *Environ Sci Technol.* **2012**;46(9):4877–4884. doi:[10.1021/es204312u](https://doi.org/10.1021/es204312u)
- [96] Jahangir D, Oh HS, Kim SR, et al. Specific location of encapsulated quorum quenching bacteria for biofouling control in an external submerged membrane bioreactor. *J Memb Sci.* **2012**;411–412:130–136. doi:[10.1016/j.memsci.2012.04.022](https://doi.org/10.1016/j.memsci.2012.04.022)
- [97] Köse-Mutlu B, Ergön-Can T, Koyuncu İ, et al. Quorum quenching MBR operations for biofouling control under different operation conditions and using different immobilization media. *New pub: Balaban.* **2015**;57(38):17696–17706.
- [98] Cheong WS, Kim SR, Oh HS, et al. Design of quorum quenching microbial vessel to enhance cell viability for biofouling control in membrane bioreactor. *J Microbiol Biotechnol.* **2014**;24(1):97–105. doi:[10.4014/jmb.1311.11008](https://doi.org/10.4014/jmb.1311.11008)
- [99] Xiao Y, Waheed H, Xiao K, et al. In tandem effects of activated carbon and quorum quenching on fouling control and simultaneous removal of pharmaceutical compounds in membrane bioreactors. *chem.Eng. J.* **2018**;341:610–617.
- [100] Lee SH, Lee S, Lee K, et al. More efficient media design for enhanced biofouling control in a membrane bioreactor: quorum quenching bacteria entrapping hollow cylinder. *Environ Sci Technol.* **2016**;50(16):8596–8604. doi:[10.1021/acs.est.6b01221](https://doi.org/10.1021/acs.est.6b01221)
- [101] Nahm CH, Choi DC, Kwon H, et al. Application of quorum quenching bacteria entrapping sheets to enhance biofouling control in a membrane bioreactor with a hollow fiber module. *J Memb Sci.* **2017**;526:264–271. doi:[10.1016/j.memsci.2016.12.046](https://doi.org/10.1016/j.memsci.2016.12.046)
- [102] Nabi M, Liang H, Zhou Q, et al. In-situ membrane fouling control and performance improvement by adding materials in anaerobic membrane bioreactor: A review. *Sci Tot Environ.* **2023**;865:161262. doi:[10.1016/j.scitotenv.2022.161262](https://doi.org/10.1016/j.scitotenv.2022.161262)
- [103] He Z, Lan X, Hu Q, et al. Antifouling strategies based on super-phobic polymer materials. *Prog Org Coat.* **2021**;157:106285.
- [104] Fitridge I, Dempster T, Guenther J, et al. The impact, and control of biofouling in marine aquaculture: a review. *Biofouling.* **2012**;28:649–669. doi:[10.1080/08927014.2012.700478](https://doi.org/10.1080/08927014.2012.700478)
- [105] Li X, Sotto A, Li J, et al. Progress and perspectives for synthesis of sustainable antifouling composite membranes containing in situ generated nanoparticles. *J Membr Sci.* **2017**;524:502–528. doi:[10.1016/j.memsci.2016.11.040](https://doi.org/10.1016/j.memsci.2016.11.040)
- [106] Sinha S, Kumar R, Anand J, et al. Nanotechnology-based solutions for antibiofouling applications: An overview. *ACS Applied Nano Materials.* **2023**;6:12828–12848. doi:[10.1021/acsanm.3c01539](https://doi.org/10.1021/acsanm.3c01539)
- [107] Soon ZY, Jung JH, Jang M, et al. Zinc pyrithione (ZnPT) as an antifouling biocide in the marine environment—a literature review of its toxicity, environmental fates, and analytical methods. *water, Air. & Soil Pollut.* **2019**;230:1–18. doi:[10.1007/s11270-018-4051-3](https://doi.org/10.1007/s11270-018-4051-3)